



CS315 – Network Security

Assignment No.1 Solution

Semester: FALL 2025

Total Marks = 20

Deadline
04th of Nov 2025

Please carefully read the following instructions before attempting the assignment Solution.

NOTE

Don't copy-paste the same answer.

Make sure you can make some changes to your solution file before submitting copy paste solution will be marked zero.

If you found any mistake then correct yourself and inform me.

Before submitting an Assignment GDB checks your assignment requirement file.

For any query, feel free to Contact at
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Provide by M.JUNAID QAZI

Question No.1

During a training session, a Security Analyst from a renowned security training company shared several real-world cases that highlighted weaknesses in organizational information security practices. According to the analyst, each incident resulted from the absence of essential M. services, as defined by the X.800 security architecture.

Read the following five cases carefully. For each case, identify the X.800 security service that is mainly compromised:

- a) A company's executive sends an email containing the strategic plan. During transmission, a hacker intercepts and reads the contents, but does not alter the message in any way.

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- b) An accountant logs into a secure system and sends a digital approval for a high-value fund transfer. The system correctly verifies his identity at login, but it does not generate any verifiable record or digital signature of the approval itself. Later, the accountant denies having sent the approval.
- c) A hospital's system sends a patient's lab test results electronically to a specialist. During transmission, the blood test values get altered due to an anonymous reason. The specialist receives the results from the hospital's verified system, but the values have been tampered.
- d) An employee successfully logs into company's system and attempts to open a restricted engineering design folder on the system. Due to a misconfigured permission setting, the system allows the employee to access the folder, even though he does not have authorization. However, the employee does not share, alter, or disclose the documents.
- e) An online banking system allows users to log in with a username and password. Due to a system flaw, the password check is bypassed, so an attacker who knows a valid username can log in without providing the correct password. Once logged in, the attacker has normal account access permissions.

Note: You need to write your answer in the following table:

Case Reference	Compromised Service
a)	Confidentiality
b)	Non-Repudiation
c)	Integrity
d)	Access Control
e)	Authentication

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Question No.2

Scenario:

A financial software system is used by a central bank to transmit secure codes for authenticating transactions between branches. In one such communication, a sender needs to transmit the numeric message “89” securely to a receiver using a Stream Cipher. The system uses a shared key stream for encryption and decryption. For this transmission, the agreed key stream is:

Key Stream: 11001010

Using the Stream Cipher process, you are required to:

- Perform the Encryption of the message “89” at the sender’s side.
- Perform the Decryption of the resulting ciphertext at the receiver’s side to recover the original message.

Note: Your answer must be in the following table:

Encryption (Sender Side)	Decryption (Receiver Side)
Plaintext “89” → Binary: 00111000 00111001Key Stream: 11001010Ciphertext (XOR result): 11110010 11110011 → (öö)	Ciphertext: 11110010 11110011Key Stream: 11001010Recovered Plaintext: 00111000 00111001 → “89”

**Every Assignment/GDB is change due to unique Student ID so don't copy
That is truly perfect step by step idea solution get help easily.**

Wish you the very best of Luck!

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